

SOLUTION ARCHITECTURE

EDUGENIE – AI-POWERED LEARNING ASSISTANT

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LAYER 1 – PRESENTATION

HTML and CSS provide the EduGenie web interface. Jinja2 supports HTML template rendering through the FastAPI application.

LAYER 2 – APPLICATION / API

FastAPI acts as the central communication layer and exposes separate functionality for the learning modules.

LAYER 3 – LEARNING MODULES

- qna.py
- explanation_module.py
- quiz_module.py
- summary_module.py
- learning_path.py

LAYER 4 – AI INTEGRATION

gemini_client.py handles Gemini configuration and communication.

LAYER 5 – AI MODELS

- Gemini 3.5 Flash – cloud-based generation.
- LaMini-Flan-T5-783M – local explanation generation.

ARCHITECTURE FLOW

Browser



HTML/CSS Interface



FastAPI / Uvicorn



Selected Module



Gemini Client / Local Model



Generated Output



FastAPI



Browser

DESIGN PRINCIPLE

Responsibilities are separated so that each major learning capability can be developed and maintained independently.